The Mathematical Limits of AI Reliability

Inevitable Failure under Strong Causality

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November 2026

Abstract

Systems that report on their own success fail characteristically: they claim a scope they did not deliver in the act of stopping. We study this failure in **strong causal environments**, where claim and fact are separated by a physical transition the model can trigger but not edit; same-origin evaluation cannot detect it.

We separate what is arithmetic from what is a claim about the world. When evaluator and policy read one observation and share one prior, the verdict is a measurable function of that pair, so $I(V; T \mid O, \Pi) = 0$ follows from how the evaluator is wired rather than from any property of evaluation; we state it as an identity and do not argue from it. The statement with content is what remains once the prior is removed from the conditioning set: on the part of the distribution where the record does not cover the prior's support, the verdict cannot lower the error of a decision that already sees the record by more than $\eta(\mathrm{cov}(o))$, with $\eta(0) = 0$. In-distribution, where the prior is informative, η is large and the bound is silent — and there the verdict can and does correlate with the truth. The bound bites on the tail, and the tail is where the tasks are worth automating. No retraining of the evaluator moves it. Three things do: coverage, a record rich enough to determine the outcome, and an added channel to the world.

We measured an evaluator with zero mutual information with the truth—this is a calculation, not an estimate. Across ten trajectories, four failing, the untrained base model answered `PASS` ten times. A constant carries no bits, so the information is zero by the definition of entropy, with no asymptotics and no independence assumption.

Two further results locate the consequence: a per-step false-completion rate bounds the probability of at least one undetected failure along a conjunctive chain, exponentially only under step independence, which we refute by counterexample; and a finite trust chain whose transitions cannot read the truth has no repair mechanism from inside, and under a separate erosion assumption (H2) — an empirical assumption, measurable and not universal — its healthy states are transient, so failure is not corrected but outlived. An external physical witness is the only intervention that adds a channel rather than improving one; its value is bounded by what the witness itself knows.

1. Introduction

There is a class of tasks where "did the model succeed?" has an answer independent of the model's opinion, and where being wrong is costly. We call these **strong causal environments**: claim and fact are separated by a physical transition the model can trigger but cannot edit. Code either runs or raises~[3]; a braking distance is either shorter than the obstacle or it is not. What matters there is not fluency or likelihood, but whether the world moved and whether the system noticed.

The failure we study is not hallucination in the ordinary sense. A model that invents a citation is wrong about a text. A model that reports `PASS` on a build that did not compile is wrong about the world, and the wrongness is invisible from inside the loop that produced it. We call this a **false completion**: the system claims a scope it did not deliver, in the same act by which it decides to stop. Its danger is independent of capability, because it terminates the process that could have found the error. A system that keeps failing is annoying; one that declares success and halts is a liability, since no further evidence is collected.

The empirical situation is worse than the vocabulary suggests. Before any training, the base model behind a preference-tuning experiment was asked to adjudicate four trajectories that had demonstrably failed; it returned `PASS` on all four. On six succeeding trajectories it also returned `PASS`. Across ten items the evaluator was not merely optimistic, it was **constant**. We do not treat that as an anecdote: it identifies which hypothesis below is satisfied in practice, and in §2.8 it collapses one asymptotic statement into an identity. A constant carries exactly zero bits, so an evaluator that always says `PASS` has mutual information zero with the truth — not approximately, not in the limit, exactly.

1.1 Why the existing remedies do not reach the problem

Text-only preference optimisation. Direct preference optimisation~[6] reshapes a policy from pairs of preferred and dispreferred outputs. When those pairs come from the same prior being corrected — the normal case, since annotator, judge and policy share a training distribution — the optimisation moves the *distribution of the output* without changing the *relation between output and world*. Its objective is a tool-layer report: a margin, a loss curve, accuracy on held-out pairs. A report can be optimised to perfection while the behaviour it certifies is untouched. Above, the objective saturated while the grounded response was produced on zero prompts. §3.1 formalises this: an intervention on the evaluator's parameters is not an intervention on the truth, so no setting of them creates information the channel does not carry.

Variance-based monitoring. A second family watches dispersion of outputs, confidence or rewards and flags instability. This cannot work, for an elementary reason: variance is translation invariant, $\mathrm{Var}(X + c) = \mathrm{Var}(X)$. A monitor measuring spread is blind by construction to a *constant* bias, and the measured failure mode is exactly a constant. The worse the deception, the more stable it looks. Dispersion measures inconsistency, not incorrectness — noise on one side, a fixed offset that averaging never removes on the other.

Both families lack a channel to the world that is not the model's own report. That lack is our object.

1.2 What we prove

An identity, and then the theorem (Theorem 2). Under same-origin evaluation — evaluator and policy read the same observation and share a prior — the verdict is a measurable function of (O, Π), so $I(V; T \mid O, \Pi) = 0$. We state that as Proposition 7 and do not call it a result: for an evaluator wired this way it is automatic, and it is not the kind of statement a measurement can refute. The result of the section is Theorem 2, which drops the prior from the conditioning set. Under a coverage condition on the prior — (A2.2′), an empirical assumption we label as such — the error of any decision that reads the verdict is within $\eta(\mathrm{cov}(o))$ of the error of a decision that reads only the record. Inside the prior's coverage η is large and the bound says nothing; on the tail $\eta \to 0$ and the verdict's contribution vanishes. The bound is therefore about a channel where the channel matters, which is narrower than "the verdict carries no information" and is the only version we can defend.

An accumulation law (Theorem 1). If each step produces a false completion with probability at least ϵ, the probability of at least one undetected false completion over an n-step chain is at least $1 - (1 - \epsilon)^n$ — under step independence. We also prove the independence-free

bound, and give the counterexample showing the exponential rate is unavailable without it; this is the weakest of the three, and we say so in the text.

An absorption theorem (Theorem 3). Model trust in the system's own verdicts as a finite Markov chain with a bankruptcy state. If transitions do not take the truth as an input, no repair conditional on failure can exist; under a separate erosion assumption (H2), healthy states are then transient and absorption is almost sure. Neither half is proved here from first principles, and we say which is which: the chain algebra is textbook, the erosion condition is an empirical assumption, and the lemma in between is the one that says why the truth cannot be an input — the same-origin assumption again.

Together they give one conclusion with a precise scope: an external physical witness — a channel to the world that is neither the model's report nor a read-out of its prior — is the only operation that *adds* a channel to the decision. We formalise it (Definition 4), give it a value (Corollary 3), and show that value belongs to the witness alone, so improving the evaluator cannot substitute for acquiring one.

1.3 What we do not claim

We do not claim reliability is impossible. We claim it is bounded, and identify the bound and its escape. A witness is necessary for escaping it, not sufficient for reliability.

Two things do move the bound, and naming them costs us the slogan that nothing does. **Coverage** moves it: (A2.2) is a statement about which part of the distribution the record occupies, so a prior that has been trained on the tail is not uninformative on the tail, and $\eta(\mathrm{cov}(o))$ is written in the form it is written so that this can be measured rather than argued. **A record that determines the outcome** moves it too, and §3.2 measures the move: adding execution hashes and exit codes to the record raised forced-choice accuracy from 0.60 to 0.80, because a truth reading had entered O. What does *not* move the bound is the evaluator. No retraining of ϕ changes $I(V;T \mid O,\Pi)$, and none changes η; scaling the evaluator acts inside the conditioning set. So the correct statement is narrower than "scale cannot help": scale helps exactly to the extent that it changes coverage or enriches the record, and the experiments that would separate those from a genuine improvement of judgement are not in this paper.

And we present no benchmark: every number here appears once, in §3, to identify which hypothesis the measured system satisfies.

A note on the title. *Inevitable* is used throughout in its measure-theoretic sense — almost sure, with respect to the measure named in the theorem — and the extent is always written. Where a statement is a positive probability rather than a probability of one, it appears as ϵ or as a bound, never as rhetoric. The word carries no weight that the equations do not carry.

2. Definitions, the Causal Graph, and Three Theorems

2.0 Notation and the four-valued verdict

Random variables are written in upper case, their values in lower case, and p_X denotes the law of X. All entropies and mutual informations are in nats unless stated otherwise. For a joint law, $H(X \mid Y)$ is the conditional entropy and $I(X;Y \mid Z)$ the conditional mutual information; the data processing inequality is written $X \to Y \to Z \Rightarrow I(X;Z) \le I(X;Y)$ [1].

The verdict V is **four-valued**, not Boolean:

$$\mathcal{V} = \{\texttt{PASS},\ \texttt{FAIL},\ \texttt{INVALID},\ \texttt{NOT_RUN}\}. \tag{2.0.1}$$

This matters for what follows. The distinction between `FAIL` and `INVALID` is not decoration: `FAIL` asserts that a scope existed and was not delivered, whereas `INVALID` asserts that the question had no referent. Collapsing the two would make the bound below look tighter than it is. We keep all four.

symbol	meaning	type
$E = (\mathcal{S}, \mathcal{A}, P, s_0, v)$	strong causal environment	Definition 1
$\mathcal{S},\ \mathcal{A}$	state and action spaces	standard Borel
P	transition kernel $(\mathcal{S} \times \mathcal{A}) \times \mathcal{B}(\mathcal{S}) \to [0,1]$	Markov kernel
v	truth function $\mathcal{S} \times \mathcal{A} \to \{0,1\}$	measurable
Z_t	step truth $v(s_t, a_t)$	$\{0,1\}$
$T = T_n = \bigwedge_{t=1}^{n} Z_t$	chain truth	$\{0,1\}$
π_θ	policy	conditional law on $\mathcal{A}$
V_ϕ	same-origin evaluator	$\mathcal{V}$
O	observation channel	random element
Π	shared prior	random element
W	independent witness	random element
ϵ	false-`PASS` rate under failure	$(0,1]$
U	exogenous factor of the environment	random element
X_k	trust-chain state	$\{0, \dots, K\}$
$\eta(\cdot)$	prior-coverage slack	$[0,1]$-valued
Δ_{wit}	value of a witness	$\mathbb{R}_{\geq 0}$

Two conventions that are used throughout and are not re-stated each time: O denotes the whole record available to the model at decision time, not a single frame; and conditioning on O always means conditioning on the σ-algebra $\sigma(O)$, so that "given O" is a statement about information, not about a particular value.

2.1 Definition 1 — the strong causal environment

Definition 1. A strong causal environment is a tuple

$$E = \big(\mathcal{S},\ \mathcal{A},\ P,\ s_0,\ v\big), \qquad P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S}), \qquad v : \mathcal{S} \times \mathcal{A} \to \{0,1\},$$

with $(\mathcal{S}, \mathcal{B}(\mathcal{S}))$ a standard Borel space, $\mathcal{A}$ Borel, $s_0 \in \mathcal{S}$, P a Markov kernel and v measurable. A run of length n is a sequence $(s_t, a_t)_{t \geq 1}$ with $s_1 = s_0$, $a_t \sim \pi_\theta(\cdot \mid s_t)$, $s_{t+1} \sim P(\cdot \mid s_t, a_t)$.

Definition 1.1 (step truth). For $t \geq 1$,

$$Z_t := v(s_t, a_t) \in \{0,1\}.$$

Axiom A1 (non-writability). The truth is a function of the realised state and action and of nothing else:

$$Z_t \ \text{ is } \ \sigma(s_t, a_t)\text{-measurable.} \tag{A1}$$

A1 is weaker than the phrase "the environment cannot be tampered with". That phrase is an engineering claim about sandboxes and permissions; A1 is a measurable-structure claim that can be used in a proof. It has an immediate and useful consequence: because Z_t is $\sigma(s_t, a_t)$-measurable, it is conditionally independent of *every* other random element given (s_t, a_t), including the model parameters. Writing θ for those parameters,

$$Z_t \ \perp\!\!\!\perp \ \theta \ \mid \ (s_t, a_t). \tag{A1.1}$$

Definition 1.2 (task chain). A task chain of length n succeeds iff every step succeeds:

$$T_n := \bigwedge_{t=1}^{n} Z_t, \qquad \{\text{success}\} = \{T_n = 1\}. \tag{2.1.1}$$

The conjunction is the substance of "strong causality": one broken link breaks the chain. Nothing below requires the steps to be identically distributed, and nothing requires them to be independent — that distinction becomes the whole content of §2.7.

Proposition 1 (when the prior-independence axiom is free). Let $O^{\text{full}} := (s_{0:t}, a_{0:t})$ be the complete record up to t. Then T_n is $\sigma(O^{\text{full}})$-measurable, hence

$$I\big(T_n; \Pi \;\big|\; O^{\text{full}}\big) = 0 \qquad \text{for every random element } \Pi. \tag{2.1.2}$$

Proof. By Definition 1.2 and A1, T_n is a measurable function of $(s_{0:n}, a_{0:n}) = O^{\text{full}}$, so $H(T_n \mid O^{\text{full}}) = 0$ and the mutual information vanishes. □

Proposition 1 locates the difficulty precisely. When the record is complete, the assumption that the prior is uninformative about the truth (A2.2 below) costs nothing — it is a theorem. The assumption has content only for a **lossy** record, which is the realistic case: the model does not observe the full state, and the gap is filled in by its prior. This is exactly the regime in which the prior can help, and the next section states when it does not.

2.2 Definition 2 — same-origin evaluation

Definition 2. An evaluator V_ϕ is *same-origin* with a policy π_θ if there exist a random element Π and measurable maps f, g such that

$$\theta = f(\Pi), \qquad \phi = g(\Pi), \tag{D2.a}$$

and if, in addition, the evaluator reads no information channel that the policy does not read:

$$T \;\perp\!\!\!\perp\; V \;\mid\; (O, \Pi). \tag{D2.1}$$

Clause (D2.a) says the two functions are two read-outs of one prior; (D2.1) says they share one channel. Together they are the formal content of "there is no independent physical witness". (D2.1) is not an assumption about the *strength* of the evaluator. An arbitrarily large, well-trained evaluator satisfies it as long as everything it has ever seen arrived through O and its own prior.

Proposition 2 ((D2.1) is a d-separation statement). (D2.1) holds iff T and V are d-separated by (O, Π) in the graph of §2.5. In that graph the only path from T to V is $T \leftarrow U \rightarrow O \rightarrow V$, and conditioning on O blocks it.

Axiom A2.2 (prior uninformative on the tail). The distinction the axiom formalises is the epistemic one: uncertainty that further evidence can remove, as opposed to noise that it cannot~[2]. On the low-coverage event

$$\mathrm{Low}_\kappa := \big\{o : \; \mathrm{cov}(o) \le \kappa\big\},$$

where $\mathrm{cov}(o) \in [0, 1]$ is the Π-mass of the region of the prior's support consistent with the record o, the prior adds nothing to the record:

$$T \;\perp\!\!\!\perp\; \Pi \;\mid\; O = o \qquad \text{for} \quad o \in \mathrm{Low}_\kappa. \tag{A2.2}$$

The quantitative form. In practice (A2.2) is not exact, and stating it as though it were would be the single easiest way to have this paper dismissed. We therefore use the following relaxation throughout, and give the exact version only as the $\eta \equiv 0$ special case:

$$\mathrm{TV}\Big(p_{T|O=o,\,\Pi} \;,\; p_{T|O=o}\Big) \;\le\; \eta\big(\mathrm{cov}(o)\big), \qquad \eta(0) = 0, \tag{A2.2'}$$

with η non-decreasing. A2.2′ says: the amount by which the prior can improve on the record is bounded by how much of the prior's support the record occupies. On the tail, $\mathrm{cov}(o) \to 0$ and the slack vanishes with it.

Why this is the right hypothesis, and not a convenient one. Strong-causal tasks are precisely those where the prior has little to say: the braking distance on ice at an unusual load, the interaction between two drugs that were never co-prescribed, the build that fails only under an uncommon allocator. These are not adversarial constructions; they are the reason the tasks are worth automating. In-distribution, A2.2 can fail — the prior genuinely knows something — and there the bound below simply does not bind. The claim of this paper is therefore not "evaluation is always uninformative" but **"evaluation is uninformative exactly where the stakes are"**, which is a narrower and, we think, more uncomfortable statement.

2.3 Definition 3 — the optimistic prior, and the measurement that locates it

Axiom A3 (optimistic prior). There is $\epsilon > 0$ such that

$$\Pr\left[V_\phi = \texttt{PASS} \;\middle|\; Z = 0\right] \;\geq\; \epsilon. \tag{A3.1}$$

Measured instance. In the companion study, the untrained base model was asked to adjudicate ten trajectories: four that had demonstrably failed and six that had succeeded. It answered `PASS` on all ten. We use this in exactly one way, and it is worth being explicit about what that way is: it tells us that (A3.1) holds with $\epsilon = 1$, and it tells us that in the measured regime V is a **constant**. The second fact is much stronger than the first, and §2.8 turns it into an identity rather than a bound.

Proposition 3 (dispersion monitors are structurally blind). Let $M = \mathrm{Var}(V_\phi)$ be any monitor that scores the evaluator by the dispersion of its output. Then

$$M(V_\phi + c) = M(V_\phi) \quad \text{for all } c, \qquad \text{and in particular} \qquad V_\phi \equiv \texttt{PASS} \;\Rightarrow\; M = 0. \tag{2.3.1}$$

Consequently, on the set of constant evaluators, M assigns the same value to one that always says `PASS` and one that always says `FAIL`: a false completion on every failure and a false completion on none are indistinguishable to it. No threshold on a variance-based monitor separates them.

Proof. $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ and $\mathrm{Var}(c) = 0$. □

Proposition 3 is the formal counterpart of §1.1's second paragraph, and it is why the empirical observation is not a coincidence: the failure mode with the largest bias is also the one with the smallest dispersion.

2.4 Definition 4 — the independent witness

Definition 4. A random element W is an *independent witness* for the truth T relative to the record O if

$$T \;\not\perp\!\!\!\perp\; W \;\mid\; O \qquad \text{and} \qquad W \;\perp\!\!\!\perp\; \Pi \;\mid\; (T, O). \tag{D4.1}$$

The first clause says W carries information about the truth that O does not already carry. The second says W is not a read-out of the prior. Both are needed, and the second is the one that is usually skipped.

Proposition 4 (the second clause is not implied by the first). There is a witness satisfying the first clause of (D4.1) and violating the second, whose information gain is zero.

Proof. Take $W := V_\phi$ itself — the evaluator offered as its own witness. The first clause of (D4.1) then holds whenever the evaluator is at all correlated with the truth. The second fails, because W is a measurable function of (O, Π). And the value is zero: $I(W; T \mid O, V) = I(V; T \mid O, V) = 0$, since V is determined by (O, V). The same computation applies to any

second evaluator $W = V_{\phi'}$ with $\phi' = h(\Pi)$ for which $T \perp \Pi \mid O$ holds. This is the "witness that is only another opinion". □

Proposition 4 is the reason the definition is written with two clauses rather than one, and it also supplies the operational test below: a candidate witness is worth exactly its conditional mutual information given the record and the verdict, and that quantity can be estimated on held-out trajectories.

The escape criterion. For a candidate witness W, define

$$\mathrm{Esc}(W) := \Big[\, I\big(W; T \;\big|\; O, V\big) > 0 \,\Big]. \tag{2.4.1}$$

$\mathrm{Esc}(W)$ is the property that matters; it is a computable functional of the joint law, not a stipulation about how W was produced.

2.5 The causal graph, d-separation, and what an intervention can do

The assumptions of §2.1–2.4 are summarised by Figure 1. Nodes are random elements; the absence of an edge is a substantive claim.

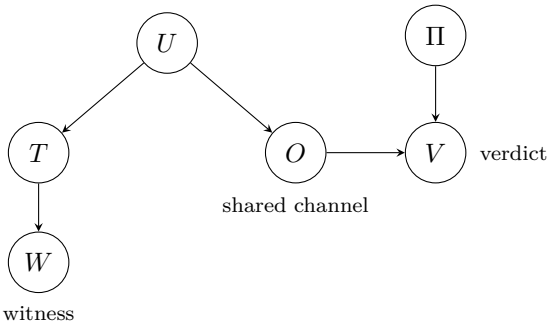


Figure 1: The causal graph of the hypotheses. U is the environment's exogenous factor and the common cause of the truth T and the record O; that edge is what makes the record informative. $O \to V$ and $\Pi \to V$ are the two inputs of a same-origin evaluator. The edge that is missing, $T \to V$, is the hypothesis (D2.1), i.e. the factorisation identity of Proposition 7. The arrow $T \to W$ is a witness: an arrow out of the truth that bypasses O.

Reading the graph:

- U is the environment's exogenous factor. It is the common cause of the truth T and the record O: this edge is what makes the record informative at all.
- $O \to V$ and $\Pi \to V$ are the two inputs of a same-origin evaluator. There is **no** edge $T \to V$ — that missing edge *is* the hypothesis (D2.1).
- $T \to W$ is the witness. A witness is, by construction, an arrow out of the truth that bypasses O.

The same graph in TikZ (the source of the figure; MathJax does not render it):

Proposition 5 (the bound, by d-separation). In the graph above, T and V are d-separated by $\{O, \Pi\}$. Hence (D2.1) holds, hence

$$I\big(V; T \;\big|\; O, \Pi\big) = 0. \tag{2.5.1}$$

Proof. The only path from T to V is $T \leftarrow U \to O \to V$. It is a chain through O, and O is in the conditioning set, so the path is blocked. The edge $T \to W$ leads nowhere, since W is not an ancestor of V. Every path is blocked, so the d-separation holds; by the vertex separation

theorem of directed graphs~[5], d-separation implies conditional independence, which is (D2.1). □

Proposition 5 is the reason the paper is organised around a graph rather than around a bare information inequality. A claim of the form "$I(V;T) = 0$" is false in general — an evaluator can and often does correlate with the truth — and a referee is entitled to reject it with a single counterexample. A claim of the form "T is d-separated from V by (O, Π)" is a statement about a mechanism, it can be checked edge by edge, and it can be broken in exactly one way.

What an intervention can and cannot do. Let $do(\phi := \phi')$ denote replacing the evaluator's parameters. Since ϕ is not an ancestor of T in the graph,

$$\Pr\left[T \;\middle|\; do(\phi := \phi')\right] = \Pr[T] \qquad \text{for every } \; \phi', \tag{2.5.2}$$

so no retraining of the evaluator moves the truth. More to the point, the *bound* is invariant as well:

$$I\big(V_{\phi'}; T \;\big|\; O, \Pi\big) = 0 \qquad \text{for every } \; \phi', \tag{2.5.3}$$

because the missing edge $T \to V$ is a property of the graph, not of the parameters. Equation (2.5.3) is the formal statement of "preference tuning cannot repair this", and §3.1 elaborates it. Conversely, $do(V := \texttt{PASS})$ — lying harder — does not change T either; you cannot improve the world by misreporting it, and you cannot learn about it by misreporting it.

The one intervention that changes (2.5.1) is structural: add the edge $T \to W$, and let the decision read W. This is what (D4.1) encodes, and it is why the witness is not a better evaluator but a second channel.

2.6 The assumption ledger (Table A0)

Every assumption the argument uses, with its kind. The kind matters more than the label, because it decides what a reader is entitled to do with a theorem that cites it. We use three kinds and no others.

- **S (structural).** Holds because of how the object is defined or wired. Not an empirical claim, and not something data can overturn. (D2.1) is of this kind: an evaluator that reads only (O, Π) is a measurable function of (O, Π).
- **M (modelling).** A choice about the model, which a different model could make differently. Asserted, not measured, and defended by the fact that the results are stated conditionally on it.
- **E (empirical).** A claim about a distribution in the world. Measurable, in principle refutable by a measurement, and **not** universal. A theorem that cites an E-assumption is a conditional statement about systems in which that measurement comes out favourably; it is not a law.

label	kind	statement	load-bearing for	if removed
A1	S	$Z_t = v(s_t, a_t)$	A1.1; Proposition 1	environment can be edited; the problem changes
A2 (D2.1)	S	$T \perp V \mid (O, \Pi)$	Proposition 7, Lemma 1	the record would not exhaust the verdict

label	kind	statement	load-bearing for	if removed
A2.2	**E**	$T \perp \Pi \mid O$ on low-coverage O	Corollary 2	only the quantitative form, Theorem 2, survives
A2.2′	**E**	TV slack $\eta(\text{cov})$	Theorem 2	the bound is vacuous, not false; only $\eta \equiv 0$ survives
A3	M	$\Pr[\texttt{PASS} \mid Z = 0] \geq \epsilon > 0$	Theorem 1	no accumulation to bound
A5	M	step independence	the *rate* in Theorem 1	linear bound only; see Proposition 6
H1	S	$\lvert\mathcal{X}\rvert < \infty$	Theorem 3	the finite-state argument
H2	**E**	erosion: every transient state has positive hazard into 0	Theorem 3, Theorem 3′	no almost-sure absorption; the chain may be irreducible
L1(i)–(ii)	M	update has no T input; X_k summarises the record	Lemma 1	marginal transition may inherit T

The two E-entries are the ones a reader should press on, and we say why. A2.2 and H2 are not axioms and we do not number them as such. Each is a claim about a distribution that a measurement could refute:

- A2.2 fails if, on records where $\text{cov}(o) \leq \kappa$, the prior still improves on the record — that is, if $\eta(\text{cov}(o))$ turns out to be bounded away from zero on the tail. §4.4 states this as the paper's first falsifier, and $0.60 \to 0.80$ in §3.2 is a partial instance of the prior improving once the record is enriched, which is why A2.2′ and not A2.2 is the form we use.
- H2 fails if trust can persist on a set of states from which bankruptcy is unreachable. A system that never loses trust because it never notices anything wrong satisfies Lemma 1 and violates H2, and Theorem 3 does not apply to it. We state this in §2.9 rather than hide it, and we do not claim that H2 is necessary for the phenomenon — only that it is the condition under which the chain argument gives absorption.

Two further entries deserve comment. A2.2′ is how we keep the argument alive when A2.2 is attacked, at the price of a slack term that is itself measurable. A5 is the weakest assumption in the paper and is discussed in §2.7, where it is also refuted in the direction that matters. A5 is labelled M rather than E because the mechanism it denies — one prior, one blind spot, positively correlated errors — is a modelling claim about how the deployment is built, and §2.7 exhibits what happens when it is false.

2.7 Theorem 1 — the accumulation law

Definition (false completion). For $t \geq 1$,

$$A_t := \{V_\phi(s_t, a_t) = \texttt{PASS}\} \cap \{Z_t = 0\}. \tag{2.7.1}$$

Axiom A5 (step independence). The events $A_1, \ldots, A_n$ are mutually independent. A5 is used only for the rate in (2.7.3); it is not needed for (2.7.2), (2.7.4) or (2.7.6), and §2.7 shows that it fails in the regime of interest.

Theorem 1. The following hold, in decreasing strength of hypothesis.

(1.1) No assumption (union bound).

$$\Pr\Big[\bigcup_{t=1}^{n} A_t\Big] \;\le\; \sum_{t=1}^{n} \Pr[A_t] \;\le\; n\,\epsilon. \tag{2.7.2}$$

(1.2) Independent steps (geometric decay). If $A_1, \dots, A_n$ are independent and $\Pr[A_t] \ge \epsilon$ for all t, then

$$\Pr\Big[\bigcap_{t=1}^{n} A_t^{\mathsf{c}}\Big] = \prod_{t=1}^{n} \big(1 - \Pr[A_t]\big) \;\le\; (1-\epsilon)^n. \tag{2.7.3}$$

(1.3) Exact criterion for the infinite product. Let $\epsilon_t := \Pr[A_t]$.

$$\prod_{t=1}^{\infty}(1-\epsilon_t) = 0 \qquad \Longleftrightarrow \qquad \sum_{t=1}^{\infty} \epsilon_t = \infty. \tag{2.7.4}$$

(1.4) Self-termination. Let $\nu := \inf\{t : V_\phi = \texttt{PASS}\}$ be the policy's own stopping rule. Then under independence

$$\Pr[\nu \le n] = 1 - \prod_{t=1}^{n}(1-\epsilon_t), \qquad \text{which tends to 1 whenever } \textstyle\sum_t \epsilon_t = \infty. \tag{2.7.5}$$

(1.5) Partial completion (a deviation bound, not a reliability bound). Let $S_n = \frac{1}{n}\sum_{t\le n} Z_t$ and $\hat{S}_n = \frac{1}{n}\sum_{t\le n} \mathbf{1}[V_\phi = \texttt{PASS}]$. Then

$$\mathbb{E}\big|\hat{S}_n - S_n\big| \;\le\; \underbrace{\Pr[V = \texttt{PASS},\, Z = 0]}_{\le \epsilon} + \underbrace{\Pr[V \ne \texttt{PASS},\, Z = 1]}_{=:\,\delta_{\mathrm{miss}}} \;\le\; \epsilon + \delta_{\mathrm{miss}}, \tag{2.7.6}$$

independently of n.

Proofs. (1.1) is subadditivity plus (A3.1). (1.2) is the product formula for independent events, plus $1-q_t \le 1-\epsilon$ when $q_t \ge \epsilon$. For (1.3), take logarithms: $\log\prod(1-\epsilon_t) = \sum\log(1-\epsilon_t)$, and use $\log(1-x) \le -x$ together with $\log(1-x) \ge -2x$ for $0 < x \le \frac{1}{2}$; the tail of the sum decides convergence, so the product vanishes iff $\sum\epsilon_t$ diverges. (1.4) is $\{\nu \le n\} = \bigcup_{t\le n} A_t$ with (1.2). For (1.5), triangle inequality on the difference of empirical means, and note that $|\mathbf{1}[V = \texttt{PASS}] - Z| \le \mathbf{1}[V = \texttt{PASS}, Z = 0] + \mathbf{1}[V \ne \texttt{PASS}, Z = 1]$. □

The assumption that does the work, and why it fails. The rate in (2.7.3) is entirely due to independence. In the systems we care about, independence is false in a specific direction: one prior, one blind spot, so errors are positively correlated. The following counterexample shows that positive correlation makes the *no-false-completion* probability larger, not smaller.

Proposition 6 (the exponential rate is not available without A5). Fix $\epsilon \in (0,1)$ and consider a chain that fails at every step, $Z_t = 0$ for all $t \le n$ — a task that cannot be done. Let the evaluator commit once per episode: draw $U \sim \mathrm{Bern}(\epsilon)$ and set $V_\phi = \texttt{PASS}$ on $\{U = 1\}$ and $V_\phi = \texttt{FAIL}$ on $\{U = 0\}$, for every t. Then A3 holds with equality, $\Pr[A_t] = \epsilon$ for every t, and

$$\Pr\Big[\bigcap_{t=1}^{n} A_t^{\mathsf{c}}\Big] = 1 - \epsilon \qquad \text{whereas} \qquad (1-\epsilon)^n \downarrow 0. \tag{2.7.7}$$

Proof. $A_t = \{U = 1\}$ for every t, so all A_t coincide and the intersection is $\{U = 0\}$. □

This is the smallest possible deviation from the hypotheses of (2.7.3): the marginals $\Pr[A_t] = \epsilon$ are exactly as required, and A3 holds with equality. Only the dependence is different, and it makes the clean-run probability *independent of* n. The counterexample therefore isolates A5

and nothing else. The interpretation is not exotic: an evaluator that forms one opinion per episode and repeats it is the normal case, not the pathological one.

Proposition 6 is not a technicality; it is a correction to how Theorem 1 may be read. **Theorem 1 must not be read as "errors accumulate".** What accumulates, under independence, is the chance of at least one false completion; under positive correlation, that chance can be smaller while each individual report is just as wrong. The quantity that degrades with n regardless of correlation is the *discrepancy between report and truth*, and that is the object of §2.8 — of Theorem 2 in particular, because the discrepancy is what the decision error is made of.

Theorem 1's honest role is therefore narrower than its statement suggests, and we prefer to say so: it is the arithmetic of a per-step false-positive rate under an assumption that real systems violate. Its useful content is (1.5): on average-scored tasks the report can be off by at most $\epsilon + \delta_{\text{miss}}$ no matter how long the chain, which is the formal reason a high single-step accuracy coexists with long-chain failure and a benchmark that looks fine.

2.8 What a verdict carries

The order of this section is deliberate. It states two propositions, four corollaries, and then Theorem 2 at the end. The first proposition is what (D2.1) yields immediately, and it is a statement about measurable functions rather than about evaluation; the second is a claim about one measured artefact. Theorem 2 is what remains when we stop conditioning on the prior, which is the situation any deployment is actually in. It comes last because it is the statement that degrades gracefully rather than failing outright when the tail assumption is weakened.

Proposition 7 (the factorisation identity). Under (D2.1), for every environment of Definition 1 and every evaluator satisfying Definition 2,

$$\boxed{I\big(V;T \;\big|\; O,\Pi\big) = 0} \tag{2.8.1}$$

with probability one, i.e. the conditional law factorises $p_{V,T|O,\Pi} = p_{V|O,\Pi}\, p_{T|O,\Pi}$.

Proof. (D2.1) *is* the factorisation; the mutual information of two random elements that are conditionally independent given Z is zero, since $I(V;T \mid Z) = D_{\mathrm{KL}}\big(p_{V,T|Z} \,\|\, p_{V|Z}p_{T|Z}\big)$ and the argument of the logarithm is 1 almost surely. □

This is not the paper's contribution, and we no longer present it as one. The proof is one line because the conclusion is contained in the hypothesis, and the hypothesis is close to automatic. An evaluator is a map from what it can read to a verdict. If what it can read is (O,Π), then V is $\sigma(O,\Pi)$-measurable, and for a measurable function of (O,Π) the conditional independence (D2.1) is not a claim about the evaluator's quality but a consequence of how it is wired. Three apparent openings close the same way. A deterministic evaluator satisfies (D2.1) by measurability. A randomised evaluator that draws independent noise satisfies it because independent noise carries nothing about T. An evaluator that reads some further channel is excluded by the second clause of Definition 2, which says in words that it reads nothing the policy does not read.

Equation (2.8.1) is therefore true, and it is the reason no amount of work on the evaluator can help, but it distinguishes no deployment from any other and no measurement can refute it. We keep it because every statement below is phrased relative to it — it fixes what "no information beyond the record" means — and because it locates the content of the paper by exclusion: the content is not in the quality of the evaluator, and it is not in (2.8.1).

Where the content is has two halves, and they must be kept apart. Off the low-coverage part of the distribution, the record and the prior may determine the truth; the verdict may then correlate with the truth through the record, and $I(V;T)$ may be large. Nothing in (2.8.1)

says otherwise, and any reading of it as "$I(V;T) = 0$" is simply false. On the low-coverage part — the tail, where strong-causal tasks live — the prior has nothing to add to the record, and there the verdict's *marginal* contribution can be shown to vanish. The first half is Corollary 1; the exact form of the second half is Corollary 2; and the form that survives an inexact tail assumption is Theorem 2, at the end of this section.

Corollary 1 (data-processing upper bound, and what it does not say).

$$I(V;T) \;\le\; I\big(V;(O,\Pi)\big) \;\le\; \min\big\{H(V),\, H(O,\Pi)\big\}. \tag{2.8.2}$$

This is an upper bound, not a vanishing quantity. $I(V;T)$ can be strictly positive, because O carries information about T; the record is informative even when the verdict is not. Any claim of the form "$I(V;T) \to 0$" is false in general and is not made here.

Inside the support of the prior this is the normal case and not a defect: the verdict inherits whatever the record knows about the truth, and $I(V;T)$ can be large. §3.2 measures the inherited part *growing* when the record is enriched — forced-choice accuracy rises from 0.60 to 0.80 once execution hashes and exit codes enter O — and every point of that gain appears on the right-hand side of (2.8.2), arriving through the record. What the verdict does not do is add to (O,Π). Corollary 1 is therefore the first half of the paper's claim: it says the verdict can be informative, and it also says that all of its information is borrowed.

Corollary 2 (posterior invariance; the exact tail case). (A2.2) is asserted only on Low_κ, so read this corollary conditioned on a record $o \in \mathrm{Low}_\kappa$: if $T \perp \Pi \mid O = o$ at that record, then for every $y \in \mathcal{V}$,

$$\Pr\big[T = 1 \;\big|\; V = y,\, O\big] = \Pr\big[T = 1 \;\big|\; O\big], \qquad \ell(y) := \log \frac{\Pr[V = y \mid T = 1, O]}{\Pr[V = y \mid T = 0, O]} = 0. \tag{2.8.3}$$

Consequently, for any decision rule $\hat{T}$,

$$\inf_{\hat{T}(\cdot)} \Pr\big[\hat{T}(V,O) \neq T\big] \;=\; \inf_{\hat{T}'(\cdot)} \Pr\big[\hat{T}'(O) \neq T\big]. \tag{2.8.4}$$

The verdict cannot improve on the record: it is not that reading V makes the error larger, but that it cannot make it smaller. The likelihood ratio the verdict contributes is exactly 1 throughout.

Proof. From (D2.1) and $T \perp \Pi \mid O$ one obtains $T \perp (V,\Pi) \mid O$ by the contraction property of conditional independence; the conditional law of T given (V,O) therefore does not depend on y, giving (2.8.3). For (2.8.4): the direction $\le$ is immediate, since a rule that ignores V is a rule that uses (V,O); the reverse direction follows from (2.8.3): a rule that uses V has the same risk as the randomised rule that draws $V \sim p_{V|O}$ and then ignores it, and randomisation cannot beat the optimal deterministic rule on O. Both directions together give equality. □

What Corollary 2 says on the tail, in one sentence. Where (A2.2) holds exactly, the verdict is informationally dead: conditionally on the record, $I(V;T \mid O) = 0$, and the Bayes error of any decision equals that of a decision that never reads V. This is the exact half of the paper's claim, and it is exact rather than asymptotic for a reason worth stating: V is a function of (O,Π), and the hypothesis removes Π from the conditioning set, so what is left of V is a function of O alone and adds nothing to O. Corollary 2 is therefore the statement "on the tail the verdict contributes zero", and it fails only if (A2.2) fails.

Proposition 8 (the measured evaluator). If $V \equiv y_0$ almost surely, then for **every** joint law of (T,O),

$$I(V;T) = 0, \qquad I(V;T \mid O) = 0, \qquad \Pr[T = 1 \mid V] = \Pr[T = 1]. \tag{2.8.5}$$

Proof. A constant is measurable with respect to the trivial σ-algebra $\{\varnothing, \Omega\}$, so $\sigma(V)$ is trivial, $H(V) = 0$, and $I(V;T) = H(V) - H(V \mid T) = 0$. The same argument gives the conditional statements. □

Proposition 8 needs no hypothesis beyond constancy: not (D2.1), not A2.2, not A3. It applies to the measured system because the evaluator answered `PASS` on all ten trajectories, failing and succeeding alike. Unlike (2.8.1) it is a factual claim about a system, because its hypothesis is an observation rather than a consequence of how the evaluator is built: the arithmetic is trivial, and the measurement is not. For a constant evaluator the information is zero by the definition of entropy, and no asymptotics is needed to say so.

Corollary 3 (the value of a witness). Define

$$\Delta_{\text{wit}} := I\big((V,W);T \mid O\big) - I(V;T \mid O) \;\geq\; 0. \tag{2.8.6}$$

The chain rule for conditional mutual information gives, as an identity and with no hypothesis,

$$\Delta_{\text{wit}} = I\big(W;T \mid O,V\big). \tag{2.8.7}$$

The inequality that bounds this quantity comes from clause (ii) of Definition 4, not from the chain rule. Applying the chain rule to $I(W;(T,V) \mid O)$ in the other order,

$$I\big(W;T \mid O\big) = I\big(W;V \mid O\big) + I\big(W;T \mid O,V\big) - I\big(W;V \mid O,T\big),$$

and $W \perp \Pi \mid (T,O)$ together with $V = f(O,\Pi)$ gives $W \perp V \mid (T,O)$, so the last term vanishes and

$$\Delta_{\text{wit}} = I\big(W;T \mid O\big) - I\big(W;V \mid O\big) \;\leq\; I\big(W;T \mid O\big), \tag{2.8.7'}$$

with equality iff $I(W;V \mid O) = 0$. Under (A2.2) the equality holds as well, and we can say why: on Low_κ the prior adds nothing to the record, so $W \perp \Pi \mid O$ follows from $T \perp \Pi \mid O$ and clause (ii) by integrating out T; since V is a function of (O,Π), that gives $I(W;V \mid O) = 0$. Hence on the tail

$$\Delta_{\text{wit}} = I\big(W;T \mid O\big) \qquad \text{exactly,} \tag{2.8.7"}$$

and the value of a witness is what the witness itself knows about the truth beyond the record — no discount term, and no term in ϕ. Off the tail the inequality in (2.8.7′) is what survives, and it is strict only when the candidate witness is correlated with the verdict given the record.

Two readings of (2.8.7′)–(2.8.7″) are worth stating, because one of them is the usual mistake. A witness that is independent of the prior given the truth and the record — clause (ii) of Definition 4 — attains the bound, and on the tail attains it exactly. A witness that is merely a second opinion from the same prior violates $W \perp V \mid (O,T)$, has $I(W;V \mid O) > 0$, and drives Δ_{wit} to zero; that is Proposition 4, recovered as the equality case rather than as a separate argument.

Corollary 4 (an error-rate floor). For $T \in \{0,1\}$ and any estimator $\hat{T}$ of T from an input X,

$$\Pr[\hat{T} \neq T] \;\geq\; h_2^{-1}\big(H(T \mid X)\big), \tag{2.8.8}$$

where h_2 is the binary entropy. With $X = V$ and $H(T \mid V) = H(T)$ — which holds by Proposition 8 — the floor is $\min\{\Pr[T=1], \Pr[T=0]\}$, the error of guessing without looking.

Theorem 2 (the coverage bound). Throughout, $\text{cov}(o) \in [0,1]$ is the Π-mass of the region of the prior's support consistent with the record o, and $\eta : [0,1] \to [0,\infty)$ is non-decreasing with $\eta(0) = 0$. Assume (A2.2′), restated here for reference:

$$\text{TV}\Big(p_{T \mid O=o,\Pi}\,,\; p_{T \mid O=o}\Big) \;\leq\; \eta\big(\text{cov}(o)\big) \qquad \text{for every } o.$$

Then for every o with $\Pr[O=o] > 0$ and every decision rule $\hat{T}(V,O)$ there is a rule $\hat{T}'(O)$ reading the record alone with

$$\Pr\big[\hat{T}'(O) \neq T \;\big|\; O=o\big] \;\leq\; \Pr\big[\hat{T}(V,O) \neq T \;\big|\; O=o\big] + \eta\big(\text{cov}(o)\big), \tag{2.8.9}$$

and consequently, averaging over O,

$$\inf_{\hat{T}'(\cdot)} \Pr\big[\hat{T}'(O) \neq T\big] \;\leq\; \inf_{\hat{T}(\cdot)} \Pr\big[\hat{T}(V,O) \neq T\big] + \mathbb{E}\big[\eta(\mathrm{cov}(O))\big]. \tag{2.8.10}$$

Proof. Fix o. By the coupling characterisation of total variation, (T,Π) and (T',Π) can be coupled on one probability space with $\Pi \sim p_{\Pi|O=o}$, $T \sim p_{T|O=o,\Pi}$, $T' \sim p_{T|O=o}$, and $\Pr[T \neq T' \mid O=o] \leq \eta(\mathrm{cov}(o))$. Draw V from $p_{V|O=o,\Pi}$, which is unaffected by substituting T' for T since it conditions on Π and o only. For any decision $y \mapsto \hat{T}(y,o)$,

$$\Pr[\hat{T}(V,o) \neq T \mid O=o] \;\geq\; \Pr[\hat{T}(V,o) \neq T' \mid O=o] - \Pr[T \neq T' \mid O=o],$$

because the event $\{\hat{T} \neq T\}$ contains $\{\hat{T} \neq T'\} \cap \{T = T'\}$. The second term is at most $\eta(\mathrm{cov}(o))$, and since T' has law $p_{T|O=o}$ the first is at least the best error achievable at o by a rule that reads o alone. This is (2.8.9). For (2.8.10), average (2.8.9) over o against p_O; the infimum on the right is attained pointwise in o by a rule the risk function makes measurable, the risk being continuous in the decision and the decision space standard Borel. □

Reading Theorem 2, and what it deliberately does not say. The theorem has two regimes, and it is a statement about only one of them.

Off the tail. If $\mathrm{cov}(o)$ is near 1, $\eta(\mathrm{cov}(o))$ may be near 1 as well and the bound is vacuous. That is the correct outcome and not a weakness: where the prior does cover the record, the prior is informative, the verdict may correlate with the truth well beyond what (2.8.9) allows, and Corollary 1 already said so. A theorem that claimed otherwise would be false, and §3.2 measures the opposite direction — enriching O raised forced-choice accuracy from 0.60 to 0.80.

On the tail. If $\mathrm{cov}(o) \to 0$ then $\eta \to 0$ and the verdict's contribution to the error of the best decision vanishes. In the exact case $\eta \equiv 0$ this is Corollary 2, i.e. $I(V;T \mid O) = 0$. The quantitative form exists because we do not believe $\eta \equiv 0$; it says that if the tail assumption is wrong, it is wrong by $\eta(\mathrm{cov}(o))$ and by nothing else.

So the claim of the paper is not "the verdict carries no information". It is: the verdict carries no information *where the record does not determine the truth*, and the amount by which that claim can fail is the same quantity that measures how thin the record is. That is why Theorem 2, and not (2.8.1), is the result of this section, and why the assumption it rests on — (A2.2′), a claim about coverage — is labelled empirical in Table A0 rather than promoted to an axiom.

What Theorem 2 is a ceiling on. It is a ceiling on a component, not on a system. A system whose record already determines the outcome is not constrained by it at all: the rule $\hat{T}'(O)$ is then as good as any rule, and the bound is satisfied trivially. The constrained case is a system that needs its verdict because the record is thin — and that is exactly the case in which the record is thin. The two conditions cannot be separated, which is the content of the parenthetical in §4.3: a system cannot be simultaneously dependent on its verdict, operating where its prior has no coverage, and reliable.

2.9 Theorem 3 — trust and its absorption

The chain. Trust in an automated system's own verdicts is the classical instance of the failure this section models~[4]. Let trust be a finite-state Markov chain $(X_k)_{k\geq 0}$ on $\mathcal{X} = \{0,1,\ldots,K\}$ with 0 the bankruptcy state, absorbing. Write the transition matrix in block form

$$M = \begin{pmatrix} 1 & \mathbf{0} \\ R & Q \end{pmatrix}, \tag{2.9.1}$$

where Q is the transient block on $\{1,\ldots,K\}$ and R the block of transitions into 0. Let $\tau_0 := \inf\{k \geq 0 : X_k = 0\}$.

Theorem 3. Assume (H1) $\mathcal{X}$ is finite, and (H2) for every $i \in \{1, \dots, K\}$ there is $m_i \geq 1$ with $\sum_{j=1}^{K} (Q^{m_i})_{ij} < 1$. Then

$$\boxed{\Pr[\tau_0 < \infty] = 1} \qquad \text{and} \qquad \mathbb{E}[\tau_0] = (I - Q)^{-1}\mathbf{1} < \infty. \tag{2.9.2}$$

(H2) is an **empirical assumption** in the sense of Table A0, not an axiom, and the difference is not bookkeeping. It asserts that trust erodes for reasons of its own — accumulated unexplained discrepancy — at a rate that does not vanish, and no measurement in this paper establishes it. The proof below is therefore the short step: it is the textbook algebra that turns (H2) into absorption. The step that connects (H2) to same-origin evaluation is Lemma 1, which follows the proof, and what it shows is narrower than it looks — that (H2) cannot be defeated from inside the system, not that (H2) holds.

Proof. Put $M_0 := \max_i m_i$. By (H2), $\sum_j (Q^{m_i})_{ij} < 1$ for each i, and since $Q^{M_0} = Q^{m_i} Q^{M_0 - m_i}$ with Q substochastic, row sums do not increase, so $c := \max_i \sum_j (Q^{M_0})_{ij} < 1$. Then $\Pr[\tau_0 > kM_0] \leq c^k \to 0$, which gives the first claim. Also $\rho(Q) \leq \|Q^{M_0}\|_\infty^{1/M_0} < 1$, so $\sum_{m \geq 0} Q^m = (I - Q)^{-1}$ converges and the expectation is the row sum of the fundamental matrix. □

Theorem 3 as stated is a textbook result, and we do not present it as a contribution. What we present is the lemma that makes (H2) *credible* rather than defeatable — and that lemma is where the same-origin assumption re-enters. No statement below upgrades (H2) to a theorem, and the conclusion of this section is conditional on it wherever it is stated.

Lemma 1 (without a witness, the truth is not an input to the chain). Assume **(i)** the update is a function of the record, the prior and the verdict and of nothing else, $X_{k+1} = h(X_k, O, \Pi, V_\phi)$; and **(ii)** the state is a sufficient summary of the record, $T \perp (O, \Pi, V_\phi) \mid X_k$.

Then the transition law of X does not depend on T:

$$\Pr\left[X_{k+1} = j \;\middle|\; X_k = i,\ T = t\right] = \Pr\left[X_{k+1} = j \;\middle|\; X_k = i\right] \qquad \text{for } t \in \{0, 1\}. \tag{2.9.3}$$

Proof. X_{k+1} is $\sigma(X_k, O, \Pi, V_\phi)$-measurable. By (ii), T is conditionally independent of (O, Π, V_ϕ) given X_k, hence conditionally independent of any measurable function of them. □

Hypothesis (i) is the substantive one and it is worth naming plainly: the truth T is not an argument of the update. A system that could condition its trust update on T would be one that observes its own failure directly — that is, one that already possesses a witness, which is exactly the case we are excluding. Hypothesis (ii) says the trust state remembers enough of the record that the remaining variables add nothing; when (ii) fails, the *marginal* transition can inherit a dependence on T through the correlation between T and O given X_k. We list (ii) in the ledger rather than assume it silently, because it is the point at which the Markov abstraction is doing real work.

Equation (2.9.3) is the whole point, but its role is narrower than it first appears, and we state the limit explicitly. It rules out **one** mechanism: a transition that fires when $T = 0$, that is, a transition of the form "detect the failure, retreat to a healthy state". It does **not** by itself force absorption. A chain whose transitions ignore T can perfectly well be irreducible on the healthy states and never absorb: a system that never loses trust because it never notices anything wrong is consistent with (2.9.3).

Absorption therefore needs the separate modelling claim (H2): that trust erodes for reasons of its own — accumulated unexplained discrepancy — at a rate that does not vanish. Lemma 1 is what makes (H2) *credible* rather than what implies it. Its content is that (H2) cannot be defeated by the system repairing itself, because every repair mechanism conditional on the truth has been excluded. The asymmetry is the point: nothing can rescue the chain from inside, and nothing outside is listening.

Theorem 3′ (two regimes, and what a witness actually changes). Split the bankruptcy hazard into an internally detectable part and an externally discovered part:

$$q_{i,0} = \underbrace{q_i^{\text{end}}}_{\text{detectable inside}} + \underbrace{\lambda\,\mathbf{1}[T=0]}_{\text{discovered outside, rate } \lambda} . \tag{2.9.4}$$

Without a witness, λ is **exogenous**: it is the rate at which the world happens to notice, and the system neither controls it nor learns from it. Conditional on the truth being failure, $T = 0$ throughout, the exit hazard from every transient state is at least λ, so

$$\Pr[\text{eventual bankruptcy} \mid T \equiv 0] = 1, \qquad \mathbb{E}[\tau_0 \mid T \equiv 0] \;\le\; 1/\lambda, \tag{2.9.5}$$

with equality iff $q_i^{\text{end}} = 0$ for every i — that is, in the pure exogenous regime where the system has no internal route to noticing anything. Every internally detectable route shortens the expectation; none lengthens it, because the exogenous hazard is still running underneath. We write $\le$ rather than $=$ because the equality is a claim about a system with no self-detection at all, and the systems we discuss have some.

A witness does not drive this probability to zero. It replaces the exogenous λ by an endogenous transition: $q_{i,0}$ becomes T-measurable and can be fed back, so the healthy set may acquire a closed class and the absorbing state becomes avoidable rather than certain.

We prefer 3′ to 3 in application, because 3′ states what a witness does — it turns an uncontrollable rate into a controllable one — instead of claiming a guarantee that no engineering artefact can provide.

2.10 What this chapter does not establish

Three boundaries, stated here so that they do not have to be discovered by a referee.

The bound is about a component, not a capability. (2.8.1) says the verdict carries no information beyond (O, Π), and (2.8.9) says how much the verdict can be worth where the prior is thin. Neither says the system cannot succeed; a system can succeed reliably on the strength of its observation alone, with an evaluator that is useless and ignored. In that case $\mathrm{cov}(o)$ is large, η is large, and Theorem 2 is silent — by design.

Necessity is not sufficiency. Corollary 3 gives the value of a witness as an upper bound, and (D4.1) gives the conditions under which it is positive. A positive value is necessary for escaping (2.8.1). It is not sufficient for reliability, which additionally requires that the witness be read correctly and that the decision rule use it.

Theorem 1 is the weakest of the three and is labelled as such. Its rate needs independence, Proposition 6 shows independence fails in the relevant direction, and the true content of the section is (1.5) plus the observation that the reliability question belongs to Theorem 2. A reader who discards Theorem 1 and keeps Theorems 2 and 3 loses nothing essential to the argument.

One assumption is empirical and untested by us. (A2.2), and therefore its relaxation (A2.2′), is a claim about coverage that this paper does not measure. Chapter 3 measures a related quantity — the prior improving once the record is enriched — but it does not measure $\eta(\mathrm{cov}(o))$ on a real tail. Until someone does, Theorem 2 is a conditional statement, and we would rather say so here than have a referee say it for us.

3. Results: the theory applied to a measured system

This paper contains no simulation of its own. Section 3 is instead an exercise in identification: for each theorem, which entity in a system that actually ran plays the role that the theorem

names, and is the hypothesis satisfied there? A theorem is not confirmed by finding an instance that satisfies its hypothesis — that would be circular. What the instance can do is show that the hypothesis is *satisfiable by a real artefact*, that the conclusion is therefore not vacuous, and that the assumptions are neither so strong as to be unattainable nor so weak as to be automatic.

All figures below come from the companion study. We restate them in the terms of §2 rather than citing them as evidence for a conclusion, and we name their confounds. Each of the three subsections is a mapping of the form "theorem → measured quantity → what the measurement can and cannot support".

3.1 Proposition 8 in a measured instance: the constant evaluator

The measurement. An untrained base model was presented with ten completed trajectories from a code-execution sandbox, four of which had demonstrably failed (the artefact did not run) and six of which had succeeded. It was asked, per trajectory, to adjudicate. It returned `PASS` on all four failures and `PASS` on all six successes: ten answers, one value.

The mapping. Definition 2 is satisfied in its degenerate form. The judge and the generator are the same untrained weights, so $\theta = \phi$ and the common prior is the entire parameter vector; the judge read no channel the generator did not read. Hypothesis (D2.1), $T \perp V \mid (O, \Pi)$, therefore holds not by argument but by construction. Axiom A3 holds with $\epsilon = 1$: on the failing items the false-`PASS` rate is 4/4.

The result, and why it is a computation. On the measured instance the empirical law of V is the point mass at `PASS`. Proposition 8 then applies with no hypothesis other than constancy:

$$I(V;T) = 0, \qquad I(V;T \mid O) = 0, \qquad \Pr[T = 1 \mid V] = \Pr[T = 1]. \tag{3.1.1}$$

The proof is one line — $\sigma(V)$ is trivial, so $H(V) = 0$ and $I(V;T) = H(V) - H(V \mid T) = 0$ — and it involves no limit, no estimator, no independence assumption, and no appeal to (D2.1) or A2.2. This is the sense in which the one statement of §2.8 that is arithmetic rather than empirical is arithmetic: for a constant, the mutual information is zero by the definition of entropy, and an estimator would only be needed if the evaluator were *not* constant.

A numerical consistency check that could have failed. The same untrained model scored 0.60 on forced-choice adjudication of the ten items. Under (3.1.1) that number is not a capability measurement at all: an evaluator that always answers `PASS` is correct exactly on the successful items, so four certifications of a failure count as errors and six of a success count as hits — six of ten, 0.60. Corollary 4 puts the floor for any predictor of T from V at $\min\{\Pr[T = 1], \Pr[T = 0]\} = 0.40$, and the observed 0.60 sits exactly at the value that constancy predicts, not somewhere above it. Had the model scored 0.75 or 0.90, Proposition 8 would have been false of this instance and the mapping in this subsection would collapse. It did not.

What this does and does not support. Constancy is established on ten items, not on the population; the identity (3.1.1) is exact for the measured instance, and for the population it is exact conditional on the evaluator remaining constant. We do not claim the latter, and Proposition 8 does not need it: the identity holds whenever the evaluator is constant, and Theorem 2 does not need constancy either — it needs (A2.2′) only, so a varying evaluator is inside its scope. The measurement's role is narrower and, we think, sharper: it exhibits a real artefact at the extreme point of the section's range, where the information is zero by constancy rather than by a limit.

The confound, stated plainly. The 4/4 is the model's forced-choice distribution over two verdict tokens, not emitted prose. Asked to generate freely, the same model committed to no verdict at all. The optimism lives in the prior's marginal over verdict tokens, not in the transcript. A reader who takes 4/4 as evidence about generated text would be over-reading it.

3.2 Corollary 1 in an intervention: preference tuning moves the marginal, not the graph

The measurement. Text-only direct preference optimisation was run on pairs constructed from surface contrast rather than from an observed effect. The objective saturated: reward accuracy 1.000 by step 60, with the loss margin reporting success. On held-out prompts the grounded response — the one whose claim matched an executed artefact — was produced on 0 of them. A later replication with 100 pairs, in which every pair carried an execution hash, an exit code and an exception string captured by a separate sidecar, moved the conditional decision but not the output contract: on ten live held-out trajectories, forced-choice accuracy rose from 0.60 to 0.80, false completions fell from 4/4 to 1/4, and free-form generation emitted no verdict marker at all.

The mapping. The first run is an instance of equation (2.5.3). The intervention $do(\phi := \phi')$ changes the evaluator's parameters and leaves the truth untouched, because ϕ is not an ancestor of T in the graph of §2.5; and the bound is invariant under the same intervention,

$$I\big(V_{\phi'}; T \mid O, \Pi\big) = 0 \qquad \text{for every } \phi'. \tag{3.2.1}$$

The measured outcome is the predicted one, and it is as strong as the prediction allows: saturating the objective to 1.000 produced grounded output on 0 prompts. A quantity that can be driven to its optimum while the behaviour it certifies does not move is, by (3.2.1), not a measurement of that behaviour. This is the formal content of the "DPO paradox" and the reason we treat the loss margin as a tool-layer report.

The second run is the more informative one, and it cuts against a naive reading. Adding execution hashes and exit codes to the *training pairs* raised forced-choice accuracy from 0.60 to 0.80 and cut false completions from 4/4 to 1/4. If the bound were a claim that no data can help, this would refute it. It is not, and the distinction is the whole content of Corollary 1.

What the annotated pairs did is put a truth reading into the record: the hashes and exit codes are functions of the executed artefact, so they belong to O and they raise $I\big(V; (O, \Pi)\big)$. By (2.8.2) this can raise $I(V; T)$, and it did — the model became measurably better at *reading* a trajectory. Nothing about (2.8.1) is touched, because that equation conditions on O and the improvement was an improvement *inside* the conditioning set. It is also the regime in which Theorem 2 is not binding: the record, not the verdict, is doing the work, and a bound on the verdict's marginal contribution says nothing against a gain that arrives through the record. This is the one direction in which the theory permits an evaluator to improve, and the measurement lands in it exactly.

What the annotated pairs did not do is change the output contract. Free-form generation still emitted no verdict marker on any of the ten trajectories. The gain lived in a discriminative head's marginal over two tokens and did not transfer to the behaviour that ships. In the language of §2: $p_{V|O,\Pi}$ improved while the deployed loop still contained no edge $T \to W$, so the hazard λ of (2.9.4) remained exogenous at run time. Training on a witness is not deploying one.

The honest reading of 0/10. Silence is not reliability. A system that emits no verdict has a false-completion rate of zero and a completion rate of zero: it has moved from claiming falsely to declining to claim, and the utility of the second is not obviously higher than the first. The measurement shows that preference tuning moved the model along the axis of *willingness to speak*, not along the axis of *knowledge of the world*. We record it as a null result on the axis that matters, and note that $0.60 \to 0.80$ on ten items with three discordant pairs is not separable from noise; the decisive test remains open.

3.3 Theorem 3 in an event: the absorption at cycle 107

The measurement. In a sandbox with a fixed capacity trajectory and a fixed shock schedule, two arms were run on the same population, seed and schedule, differing only in how a cycle is booked: against the agent's own report, or against an independent reading of the post-state taken by a separate instrument. The `self` arm, which assessed its own work using only the record, went extinct at cycle 107: zero functional agents, no recovery. Over the run it recorded 8871.71 units of delivery and landed 7676.93, a ledger gap of 1194.78 units, or 15.563% of what it delivered. The `r7` arm, identical except for the ledger map, was not extinct at the horizon; at cycle 107 it had delivered 8892.11 against the `self` arm's 7676.93, and at the horizon its recorded output equalled its delivered output exactly, a gap of 0.00.

The mapping. The `self` arm is an instance of the hypothesis of Lemma 1: the truth T is not an argument of the trust update. Nothing in the agent's transition law reads whether the work landed, so by (2.9.3) the chain's transitions do not depend on T, and — given the erosion condition (H2) — Theorem 3 makes the healthy set transient. The 15.563% gap is not a separate phenomenon; it is the accumulated discrepancy that the absorbing chain is a model of. The two quantities are the same quantity at different resolutions: the gap is what the ledger accumulated, and the extinction is what happened when the accumulated gap met a capacity constraint.

The ablation is the interesting part. Lemma 1 says no transition conditional on T can exist. The `r7` arm is that transition, added from outside. Its effect is exactly what Theorem $3'$ predicts: the bankruptcy hazard $q_{i,0} = q_i^{\text{end}} + \lambda \mathbf{1}[T = 0]$ of (2.9.4) has its exogenous term λ replaced by a T-measurable one, so the absorbing state becomes avoidable rather than certain. The measured consequence — a 15.8% delivery advantage at the cycle where the other arm dies, and a 3.07× advantage at the horizon — is large, but the structural point is smaller and more important: *one* variable was changed, the ledger map at the moment a cycle is classified, and it alone was sufficient to remove the extinction under an unchanged capacity trajectory.

Two constraints on that reading, both from the original record. First, the `r7` arm never enters dormancy, so it cannot strand; its survival is the joint consequence of the witness and the absence of a dormant phase, and the ablation isolates the ledger map, not "witnesses cause immortality". Second, the `r7` arm reports 4950 `INVALID` cycles against 708 in the `self` arm. That is not deterioration. The channel books the part of the scope that exists, so the cycles it reclassifies leave the `PASS` column rather than the ledger; the count of `INVALID` is therefore a measure of the reclassification, and the equality of recorded and delivered output a measure of the correction.

3.4 What the three mappings establish

theorem	hypothesis	measured counterpart	verdict
Prop. 7 (2.8.1)	(D2.1), same-origin	one model judging its own output: $\theta = \phi$	holds by wiring; no empirical content
Prop. 8 (2.8.5)	V constant	`PASS` on 10/10 trajectories	satisfied; an identity (3.1.1)
Theorem 2 (2.8.9)	(A2.2′), coverage η(cov)	$0.60 \to 0.80$ once O is enriched; $4/4 \to 1/4$	predicted: η falls as the record thickens
eq. (3.2.1)	$do(\phi)$ leaves the graph	reward accuracy 1.000, grounded output 0	predicted and observed
Cor. 1 (2.8.2)	gain must arrive through O	witness in the training pairs: $0.60 \to 0.80$ forced choice	predicted and observed
(D4.1) at run time	deployed loop has no $T \to W$	free-form verdicts 0/10	violated; no transfer

theorem	hypothesis	measured counterpart	verdict
Theorem 3 (2.9.2)	Lemma 1, (H1)–(H2)	no T-input to the trust update; extinction at 107	consistent
Theorem 3′ (2.9.5)	exogenous λ	`r7` arm: λ becomes endogenous; no extinction	consistent

The second and third rows are the ones to keep apart. Proposition 8 is an identity whose hypothesis is an observation, and it is satisfied at the extreme point of the range. Theorem 2 is the quantitative statement, and its hypothesis is the one entry in the table that the measurements in this section do not settle: $0.60 \to 0.80$ is consistent with η falling as the record thickens, but it does not estimate η itself, and we do not report it as if it did.

Three things follow, and only three.

The hypotheses are satisfiable by real artefacts. (D2.1) is not a convenient fiction: a deployed system satisfied it exactly, because generator and judge were one object. This is the weakest possible way for a hypothesis to be true and the most common.

The conclusions are not vacuous. Each theorem's conclusion was observed in a system that was not built to satisfy it. That is worth stating precisely: the theorems were not fitted to these numbers, and the numbers were not collected to test these theorems.

Nothing here is a confirmation. A single model size, a single seed, three held-out prompts, ten trajectories, one capacity trajectory, one shock schedule. The data locate the assumptions; they do not establish the theorems, which stand or fall on their proofs. A reader who wants the empirical claim to be stronger than "the hypotheses are attainable and the conclusions were observed" will not find it in this section, and should not look for it in this paper.

4. Discussion

4.1 What a benchmark measures

The word "reliability" is used for two different quantities, and the confusion is not terminological. Let V be a system's answer and V' the answer its grader would give. A benchmark reports agreement,

$$\text{score} = \Pr[V = V'], \tag{4.1.1}$$

while reliability is truth,

$$\text{acc} = \Pr[V = T]. \tag{4.1.2}$$

These are unrelated orderings on the same systems. When V and V' share a channel and a prior — the grader is the same model~[7], or a model trained on the same distribution, or a rule written by the same team from the same data — the agreement can be brought arbitrarily close to 1 at any level of accuracy. $\Pr[V = V'] = 1$ is compatible with any value of $\Pr[V = T]$ whatsoever; in the measured instance the two coexisted at 1.00 and 0.60, the second being exactly the accuracy that constancy implies — the base rate of success in the sample, not one point above it.

This is the precise sense in which a benchmark measures complicity rather than capability. The score of (4.1.1) is a functional of the joint law of (V, V') and carries no term in T. It cannot, therefore, distinguish a system that reads the world from a system that reads its own prior and reports what the prior expects. On the bulk of a distribution the two are hard to separate and the distinction hardly matters. On the tail they separate completely — and the tail is where

the tasks are worth automating. A braking system is not graded on the roads it has seen a million times; it is graded on ice, at load, at night.

Corollary 1 gives the reason this is not merely a complaint about metric design. $I(V;T) \leq I(V;(O,\Pi))$: whatever a score can measure has already passed through the record and the prior. There is no term in the score that could carry information the pair did not already carry, and therefore no reweighting, no harder subset, and no larger model can turn a same-origin score into a measurement of causal capability. What such a score measures, when it is high, is the coherence of a prior with itself.

The practical consequence is uncomfortable and follows directly. A benchmark suite built by the same organisation that built the system, evaluated on data drawn from the same pipeline, adjudicated by a grader sharing the same weights, has an upper bound on what it can detect — and that bound is the information in O. On the long tail, where the record determines almost nothing, the bound is almost nothing, and a score of 1.00 there is a statement about shared blind spots.

4.2 Why the witness is not an optimisation

It would be natural to read §3.3 as an argument for a better evaluator. It is the opposite, and Corollary 3 is the reason. The value of adding a witness is

$$\Delta_{\text{wit}} = I\big(W;T \mid O,V\big) \;\leq\; I\big(W;T \mid O\big), \tag{4.2.1}$$

with equality on the tail, by (2.8.7″), and the right-hand side contains no term in ϕ. Whatever the evaluator's quality, the marginal value of the witness is capped by what the witness itself knows about the truth beyond the record. Improving V cannot raise that cap; only changing W can. The evaluator and the witness are not on a single scale of "quality", and the intuition that a sufficiently good judge would make an external check redundant is exactly the intuition Theorem 2 forbids.

This is why we describe the intervention as a rescue rather than an optimisation. An optimisation moves a system along a frontier; a rescue changes which frontier it is on. Formally, $do(\phi := \phi')$ leaves the graph of §2.5 unchanged and leaves (2.8.1) true, for every ϕ' (equation 2.5.3). Adding W changes the graph, and it is the only intervention in this paper that does. The distinction is not rhetorical: it predicts that an unbounded budget spent on evaluator quality buys nothing at the bound, while a single cheap independent reading buys $\Delta_{\text{wit}} > 0$ — and the two predictions are not the same prediction with different constants.

There is a second, less obvious effect, and it is the one §3.3 measures. In the two-regime model of Theorem 3′, the bankruptcy hazard decomposes as $q_{i,0} = q_i^{\text{end}} + \lambda \mathbf{1}[T = 0]$, where λ is the rate at which the world *happens* to notice a failure — an auditor, a customer, an outage. Without a witness, λ is exogenous: it is a property of the environment's patience, not of the system, and no internal change alters it. With a witness, the second term becomes T-measurable, and the system can act on it. The difference is not a reduction in the probability of failure; it is a change in *who owns the response*. A system whose failures are discovered only from outside has no error-correcting dynamics, only a hazard rate, and by (2.9.5) its eventual ruin is certain with expected time at most $1/\lambda$, and exactly $1/\lambda$ only if it has no internal route to noticing anything. A system that reads its own failures has a policy.

This reframes the engineering question, and we prefer the reframed version. The question is not "how accurate is our evaluator" but "what is $I(W;T \mid O,V)$, and what does a unit of it cost". The first has no answer that survives a distribution shift; the second is a measurable quantity, bounded above by (4.2.1), and it is the only one of the two that appears in the decision.

4.3 The bound, and what it does not say

We are aware of the temptation to read a limit theorem as a verdict on an industry, and we decline it. Three things are not claimed.

Not that AI systems do not work. (2.8.9) is a statement about a component, conditional on a record. Where the record is rich, the record does the work: a system whose observation determines the outcome needs no evaluator at all, and then $\eta(\mathrm{cov}(o))$ is large and the bound does not bind — which is why we write it as a bound rather than as an equality. The theorem does not say that correct behaviour is impossible; it says that the *verdict's marginal contribution* vanishes where the record is thin, which is a claim about a component and not about a system.

Not that the tail is unlearnable. A2.2 is a statement about prior coverage, not about a permanent incapacity. A prior that has seen the tail ceases to be uninformative on it, and A2.2′ makes the slack quantitative precisely so that the day it stops holding can be recognised. What the theory rules out is a system that is simultaneously (i) dependent on its own verdict for correctness, and (ii) operating where its prior has no coverage, and (iii) reliable. Pick two.

Not that a witness suffices. $\Delta_{\mathrm{wit}} > 0$ is necessary for escaping (2.8.1) and is not sufficient for reliability. A witness must be read correctly, and the decision rule must use what it reads; the `r7` arm of §3.3 is a demonstration that the channel can be built, not a proof that any channel will be used well.

What remains after these retractions is narrower than the slogan, and easier to defend. In a strong causal environment, on the part of the distribution where the prior is thin, a system that certifies its own work has a reliability ceiling that is a property of what it can see and not of how well it judges. The ceiling does not move with the quality of the self-assessment, with objective design, or with the scale of the evaluator: each of those acts inside the conditioning set and changes nothing about what is in it.

It does move, in two ways, and denying that would be overstating the result. It moves when coverage moves — a prior that has been trained on the tail is not uninformative on the tail, and $\eta(\mathrm{cov}(o))$ is written in measurable form precisely so that this can be recognised when it happens. And it moves when the record thickens enough to determine the outcome, which is what §3.2 measured when execution hashes entered O and accuracy went from 0.60 to 0.80. Neither move is an improvement of the evaluator; both change what the decision can see.

The third move is structural, and it is the one this paper is about: adding an arrow to the graph. Between the three there is no fourth. Parameters of the evaluator leave O, Π and the graph unchanged, and therefore leave η unchanged; that is the precise sense in which no budget spent on judgement buys anything here, and it is a narrower claim than saying that scale never helps.

The organisational version of this is worse than the technical version. A system that reports `PASS` without a witness does not merely fail; it destroys the information that would have allowed the failure to be attributed. The ledger of §3.3 recorded 8871.71 against a delivery of 7676.93: at the moment of extinction the record said the work had been done, and the discrepancy was recoverable only because a separate instrument had been measuring the same thing all along. Remove that instrument and the failure becomes indistinguishable from success at every point where a decision is made — by the system, and by whoever reads its reports.

4.4 What would refute this

A theory of limits should name its falsifiers, and these are ours. We list only falsifiers that a measurement could in fact produce. An earlier draft of this section opened with "a counterexample to (D2.1)", and that was a mistake: (D2.1) holds by measurability for any evaluator whose inputs are (O, Π), so no measurement can refute it, and a falsifier that cannot fail is not a falsifier. The list below is correspondingly shorter and harder to satisfy.

A measurement of $\eta(\mathrm{cov}(o))$ that stays bounded away from zero on the tail. This is

the direct refutation of (A2.2′) and therefore of Theorem 2: find a task family where the record occupied little of the prior's support, and where a decision reading the verdict beats the best decision reading the record alone by more than η. We would want the number, because it would tell us where the ceiling is rather than whether one exists. A2.2′ was written in measurable form precisely so that this could be tested rather than argued, and we would regard such a measurement on a real task as more valuable than any further theorem in this paper.

A same-origin system that survives the tail. If a system with no external witness were shown to be reliable on strong-causal tasks in the low-coverage regime, then (A2.2), the definition of `PASS`, or our account of coverage would have to be wrong about it. We would want to know which, because any of those answers would be more interesting than the theorem. Note what is *not* on this list: a system whose evaluator reads the world directly — a controller with a force sensor in the loop — is not a counterexample but a witness, and Proposition 2 says so: it has the edge out of T that the hypothesis excludes.

Data and Code Availability: Every artifact behind this paper and its companion study — the Lean 4 formalisation, the audit scripts, and the reproducibility record — is in the public repository `https://github.com/liao-zhizhi/causal-evolution-paper`. The archived version of this preprint carries DOI `https://doi.org/10.5281/zenodo.24041061`. The repository is the evidence for the one claim in this section that is not a falsifier: that every bound stated here can be re-derived and every number re-measured from sources rather than from a summary of them.

5. Conclusion

A system that grades its own homework is not measuring its homework. This is usually said as a joke about institutions; we have tried to show that the precise version of it is a statement about what a decision can see. When the evaluator and the policy read one observation and share one prior, the verdict is a measurable function of that pair, the conditional mutual information is zero, and no amount of evaluator quality moves it — because the evaluator's parameters are not ancestors of the truth. That identity is arithmetic, and we have stopped presenting it as the paper's result: it holds by wiring, and it is not the kind of statement a measurement can refute.

The result is the statement that survives dropping the prior from the conditioning set: on the part of the distribution where the record does not cover the prior's support, the verdict cannot lower the error of a decision that already sees the record by more than $\eta(\mathrm{cov}(o))$, and there the amount is estimable. In the instance we measured the extreme case was exact rather than asymptotic: a constant evaluator, ten answers, one value, zero bits.

Three consequences follow and we have proved each in its own register. Along a conjunctive chain, a per-step false-completion rate bounds the chance that at least one failure goes undetected, and the familiar exponential rate is a consequence of an independence assumption that fails in the direction that matters. In the trust dynamics, a finite chain whose transitions do not read the truth has no repair mechanism from inside; given an independent erosion assumption — empirical, measurable, and stated as such — its healthy states are transient, so failure is not corrected but merely outlived. And in both cases the intervention that changes the bound is a channel to the world that is neither the report nor a read-out of the prior.

We have not shown that current systems are worthless, and we would regard that reading as a misuse of the result. We have shown something narrower and harder to argue with: that in the environments where being wrong is expensive, the reliability of a self-certifying system is capped by what it can see, that the cap is computable, and that three things move it — coverage, a record rich enough to determine the outcome, and an added arrow. The first two are not improvements of the evaluator, and the third is not an optimisation of anything. The cap is not a prediction about the next model. It is a statement about which measurements

could tell us anything, and it implies that most of the numbers now being reported cannot.

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